

17.1 The cell-means model

With several factors the observations fall into *cells*, one for each combination of levels. The effects model of [Chapter 8](#) writes the mean of a cell as $\mu + \alpha_i + \beta_j + \gamma_{ij}$ and pays for it with side conditions and codings. The cell-means model gives each cell its own parameter and nothing else. It fits the same values as the effects model with all interactions, but every question about the factors becomes a statement about a table of means, and the counts enter only through the precision with which that table is estimated.

17.1.1 The model

Definition 17.1.1. Cell-means model

Let factors $F_1, \dots, F_k$ have $a_1, \dots, a_k$ levels. A **cell** is a combination $\omega = (i_1, \dots, i_k)$ of levels, and there are $p = a_1 a_2 \cdots a_k$ cells. Cell ω contains $n_\omega \geq 0$ observations $y_{\omega 1}, \dots, y_{\omega n_\omega}$. It is **occupied** if $n_\omega \geq 1$ and **empty** otherwise. Write O for the set of occupied cells, $m = |O|$ and $n = \sum_\omega n_\omega$. The **cell-means model** is

$$y_{\omega l} = \mu_\omega + \varepsilon_{\omega l}, \quad l = 1, \dots, n_\omega,$$

with errors that are uncorrelated with mean zero and variance σ^2 , and independent $N(0, \sigma^2)$ when we test. In matrix form $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\mu}$, where $\boldsymbol{\mu} \in \mathbb{R}^p$ collects the cell means and $\mathbf{W}$ is the $n \times p$ matrix whose column ω indicates the observations in cell ω . We write $\mathbf{D} = \mathbf{W}^\top \mathbf{W} = \text{diag}(n_\omega)$, and $\bar{y}_\omega$ for the average of an occupied cell.

Empty cells keep their entries in $\boldsymbol{\mu}$: their means do not enter the distribution of the data, but they are properties of the population ([Section 17.4](#)). In two-factor notation, cells are pairs (i, j) , counts are n_{ij} , and we order the cells row by row: $\boldsymbol{\mu} = (\mu_{11}, \dots, \mu_{1b}, \mu_{21}, \dots, \mu_{ab})^\top$. With this ordering, a Kronecker product $\mathbf{P} \otimes \mathbf{Q}$ acts on $\boldsymbol{\mu}$ by transforming the $a \times b$ table (μ_{ij}) into $\mathbf{P}(\mu_{ij})\mathbf{Q}^\top$ (Proposition 1.10.1(f) applied to the transposed table).

By Proposition 8.6.2(a), the overparameterized two-way model with interaction has column space $\mathcal{C}(\mathbf{W})$, so the cell-means model is its identifiable parameterization. It is also a one-way layout (Definition 15.1.1) whose groups are the m occupied cells. The factorial structure plays no part in fitting.

Proposition 17.1.1. Inference with cell means

In the model of [Definition 17.1.1](#), let $\mathbf{D}^+$ be the diagonal matrix with entries $1/n_\omega$ for occupied cells and 0 for empty ones, and let $\bar{\mathbf{y}} \in \mathbb{R}^p$ hold $\bar{y}_\omega$ for occupied cells and 0 for empty ones. Then:

- $\text{rank}(\mathbf{W}) = m$, and $\mathbf{M} = \mathbf{W}\mathbf{D}^+\mathbf{W}^\top$ replaces each observation by the average of its cell;
- $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$ is estimable iff $\lambda_\omega = 0$ for every empty cell. Its best linear unbiased estimator is $\boldsymbol{\lambda}^\top \bar{\mathbf{y}} = \sum_{\omega \in O} \lambda_\omega \bar{y}_\omega$, with variance $\sigma^2 \sum_{\omega \in O} \lambda_\omega^2 / n_\omega$;
- $\text{SSE} = \sum_{\omega \in O} \sum_l (y_{\omega l} - \bar{y}_\omega)^2$ has $n - m$ degrees of freedom, and $s^2 = \text{SSE} / (n - m)$, the pooled within-cell variance, is unbiased for σ^2 ;
- if $\mathbf{L}$ is a $q \times p$ matrix of rank q whose rows vanish on the empty cells, $\mathbf{d} \in \mathbb{R}^q$, and $n > m$, the F statistic for $H : \mathbf{L}\boldsymbol{\mu} = \mathbf{d}$ is

$$F = \frac{(\mathbf{L}\bar{\mathbf{y}} - \mathbf{d})^\top (\mathbf{LD}^+ \mathbf{L}^\top)^{-1} (\mathbf{L}\bar{\mathbf{y}} - \mathbf{d})}{q s^2} \sim F(q, n - m, \gamma_H), \quad (17.1.1)$$

where $\gamma_H = (\mathbf{L}\boldsymbol{\mu} - \mathbf{d})^\top (\mathbf{LD}^+ \mathbf{L}^\top)^{-1} (\mathbf{L}\boldsymbol{\mu} - \mathbf{d}) / \sigma^2$.

Proof.

- (a) The columns of $\mathbf{W}$ for occupied cells are nonzero indicators with disjoint supports, and those for empty cells are zero. So $\text{rank}(\mathbf{W}) = m$. Since $\mathbf{D}\mathbf{D}^+\mathbf{D} = \mathbf{D}$, $\mathbf{D}^+$ is a generalized inverse of $\mathbf{W}^\top \mathbf{W}$, and $\mathbf{M} = \mathbf{W}\mathbf{D}^+ \mathbf{W}^\top$ by Theorem 6.3.6. The entry of $\mathbf{W}\mathbf{D}^+ \mathbf{W}^\top \mathbf{y}$ for an observation in cell ω is $n_\omega^{-1} \sum_l y_{\omega l} = \bar{y}_\omega$.
- (b) The rows of $\mathbf{W}$ are the coordinate vectors $\mathbf{e}_\omega^\top$ of the occupied cells, so $\mathcal{C}(\mathbf{W}^\top) = \text{span}\{\mathbf{e}_\omega : \omega \in O\}$, and Theorem 8.3.1 gives the criterion. The vector $\mathbf{D}^+ \mathbf{W}^\top \mathbf{y} = \bar{\mathbf{y}}$ solves the normal equations (Theorem 6.4.2), so $\boldsymbol{\lambda}^\top \bar{\mathbf{y}}$ is the least squares estimator of an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$, and it is best linear unbiased by the Gauss–Markov theorem (Theorem 7.2.1). The cell averages are uncorrelated with variances σ^2/n_ω , which gives the variance.
- (c) $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$ is the displayed sum by (a), and $E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - m)$ by Theorem 2.3.1, since $(\mathbf{I} - \mathbf{M})\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$.
- (d) Apply Theorem 11.3.2 with $\mathbf{G} = \mathbf{D}^+$, $\hat{\boldsymbol{\beta}} = \bar{\mathbf{y}}$ and $\boldsymbol{\Lambda}^\top = \mathbf{L}$, which is testable by (b). The matrix $\mathbf{LD}^+ \mathbf{L}^\top$ is nonsingular: for $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^\top \mathbf{LD}^+ \mathbf{L}^\top \mathbf{x} = \sum_{\omega \in O} (\mathbf{L}^\top \mathbf{x})_\omega^2 / n_\omega$. If this were zero, $\mathbf{L}^\top \mathbf{x}$ would vanish on O , and hence everywhere, because the rows of $\mathbf{L}$ vanish off O . That contradicts $\text{rank}(\mathbf{L}) = q$. ■

The counts enter only through $\mathbf{D}^+$, the variances of the cell averages. An unbalanced cell-means analysis has one fitting step, averaging within cells, and one inferential tool, Eq. 17.1.1. The work lies in choosing $\mathbf{L}$.

17.1.2 Standard hypotheses as Kronecker products

Let every cell of a two-way layout be occupied. Let $\mathbf{C}_a$ be an $a \times (a - 1)$ matrix whose columns form a basis of the contrasts $\{\mathbf{u} \in \mathbb{R}^a : \mathbf{1}^\top \mathbf{u} = 0\}$, and define $\mathbf{C}_b$ in the same way. These are coding matrices in the sense of Definition 8.5.1. The $a \times a$ centring matrix of Chapter 16 has the same column space, but its columns are linearly dependent, so we keep the thinner matrix here and count restrictions by its columns. The three classical hypotheses are

$$H_A : (\mathbf{C}_a^\top \otimes b^{-1} \mathbf{1}_b^\top) \boldsymbol{\mu} = \mathbf{0}, \quad H_B : (a^{-1} \mathbf{1}_a^\top \otimes \mathbf{C}_b^\top) \boldsymbol{\mu} = \mathbf{0}, \quad H_{AB} : (\mathbf{C}_a^\top \otimes \mathbf{C}_b^\top) \boldsymbol{\mu} = \mathbf{0}. \quad (17.1.2)$$

By the remark on Kronecker products above, H_A says that $\mathbf{C}_a^\top$ annihilates the vector of **unweighted row means** $\bar{\mu}_{i\cdot} = b^{-1} \sum_j \mu_{ij}$, that is, that these means are all equal. H_B says the same of the unweighted column means. H_{AB} says that $\mathbf{C}_a^\top (\mu_{ij}) \mathbf{C}_b = \mathbf{0}$. Because $\mathcal{C}(\mathbf{C}_a) = \mathbf{1}_a^\perp$ and $\mathcal{C}(\mathbf{C}_b) = \mathbf{1}_b^\perp$, this holds iff the doubly centred table $\mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}$ vanishes, that is, iff the cell means are additive (Definition 16.2.1). The ranks are $a - 1$, $b - 1$ and $(a - 1)(b - 1)$ (Proposition 1.10.1(c)), and the hypotheses do not depend on the basis of contrasts (Exercise (B2)). They are the hypotheses tested in the balanced layout by Theorem 16.3.1, where every

type of sum of squares tests them.

17.1.3 Weighted and unweighted marginal means

A row of the table can be summarized by any average of its cells. Two choices dominate practice. The **weighted row mean** $\bar{\mu}_i^w = n_i^{-1} \sum_j n_{ij} \mu_{ij}$ weights each cell by its count. Its natural estimator is the raw average $\bar{y}_{i.}$ of all observations in row i . The **unweighted row mean** $\bar{\mu}_i$ gives each column the same weight. Its estimator is the average of the cell averages, $\tilde{y}_i = b^{-1} \sum_j \bar{y}_{ij}$. More generally, one may fix weights $w_1, \dots, w_b$ summing to one, such as the shares of the levels of B in a target population, and estimate $\sum_j w_j \mu_{ij}$.

Proposition 17.1.2. Precision of marginal means

Fix a row i with all $n_{ij} \geq 1$, and weights $w_1, \dots, w_b$ with $\sum_j w_j = 1$. The estimator $\sum_j w_j \bar{y}_{ij}$ of $\sum_j w_j \mu_{ij}$ is unbiased with variance $\sigma^2 \sum_j w_j^2 / n_{ij}$. Among all such weights the variance is smallest, and equal to $\sigma^2 / n_{i.}$, exactly for the count weights $w_j = n_{ij} / n_{i.}$. In particular

$$\text{Var}(\tilde{y}_i) = \frac{\sigma^2}{b^2} \sum_j \frac{1}{n_{ij}} \geq \frac{\sigma^2}{n_{i.}} = \text{Var}(\bar{y}_{i.}),$$

with equality iff the counts $n_{i1}, \dots, n_{ib}$ are equal.

Proof. Unbiasedness and the variance come from Proposition 17.1.1(b). By the Cauchy–Schwarz inequality,

$$1 = \left(\sum_j \frac{w_j}{\sqrt{n_{ij}}} \sqrt{n_{ij}} \right)^2 \leq \left(\sum_j \frac{w_j^2}{n_{ij}} \right) n_{i.},$$

with equality iff $w_j / \sqrt{n_{ij}}$ is proportional to $\sqrt{n_{ij}}$, that is, $w_j \propto n_{ij}$. The unweighted mean is the case $w_j = 1/b$, which is proportional to n_{ij} iff the counts are equal. ■

The count weights are the most *precise*, but precision is the wrong criterion for choosing a target. The weighted mean $\bar{\mu}_i^w$ is the mean of row i with the levels of B mixed as they happened to be mixed in this study. If the counts are an accident of the design, such as lost plots, that mixture describes nothing. If they are proportional to a real population, as in a random sample, $\bar{\mu}_i^w$ may be exactly the target. When the rows have different mixes, comparing weighted means also compares mixes of B , and that is what the Type I sum of squares of a factor entered first tests (Section 17.2). Choose the weights to describe a population, and accept the variance that follows.

17.1.4 An unbalanced field trial

Example. Three varieties at three sites

A (synthetic) trial compared three wheat varieties V1, V2 and V3 at three sites S1, S2 and S3, with yield in tonnes per hectare. Each site grew mostly the variety it already knew. There are 37 plots in the $m = 9$ cells:

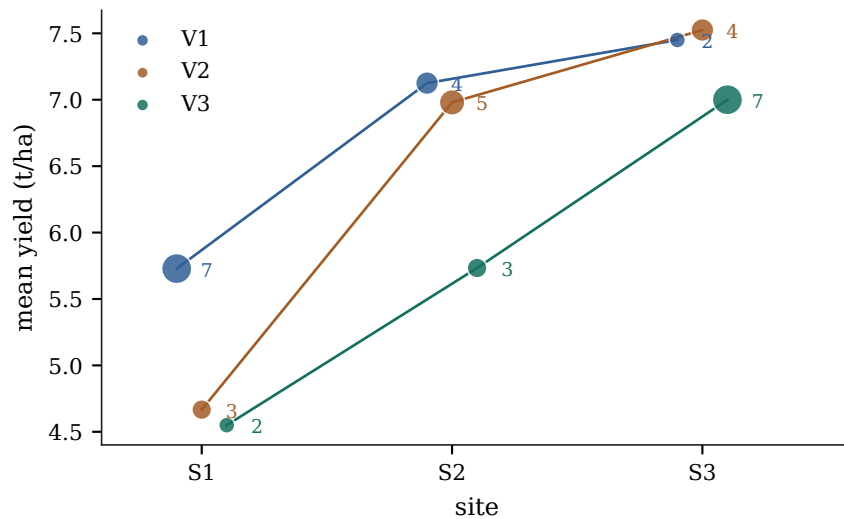


Figure 17.1.1: Cell means of the variety trial; point areas and labels give the cell counts.

	S1	S2	S3	all sites
V1	7 plots, 5.73	4, 7.12	2, 7.45	13, 6.423
V2	3, 4.67	5, 6.98	4, 7.53	12, 6.583
V3	2, 4.55	3, 5.73	7, 7.00	12, 6.275

The last column gives each variety's raw average, the estimate of its weighted mean $\bar{\mu}_i^w$. The pooled within-cell variance is $s^2 = 0.3136$ on $n - m = 28$ degrees of freedom, so the standard errors of the cell averages range from 0.212 (seven plots) to 0.396 (two plots).

Figure 17.1.1 shows the table. V1 is the best variety at S1 and S2 and nearly ties with V2 at S3, yet its raw average, 6.423, is below that of V2, 6.583, because V1 was grown mostly at the poorest site. The unweighted means 6.768, 6.391 and 5.761 compare the varieties over the same three sites, and V1 comes first. Their standard errors, 0.176, 0.165 and 0.184, exceed those of the raw averages (0.155, 0.162 and 0.162), as Proposition 17.1.2 requires. For V1 the ratio depends only on the counts: $\sqrt{13(1/7 + 1/4 + 1/2)}/3 = \sqrt{11.607}/3 \approx 1.136$.

The tests Eq. 17.1.2, computed with Eq. 17.1.1, give $F = 7.90$ on 2 and 28 degrees of freedom for equal unweighted variety means ($p = 0.0019$), $F = 44.06$ for equal unweighted site means, and $F = 1.71$ on 4 and 28 degrees of freedom for additivity ($p = 0.1755$). The difference between V1 and V2, averaged over sites without weights, is estimated as 0.377 with standard error 0.242. With $t_{28,0.025} = 2.048$, the 95% interval is from -0.12 to 0.87 . The raw averages give a difference of -0.160 , with the opposite sign.

```
import numpy as np
import pandas as pd
from scipy import stats
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
```

```
for i in range(3) for j in range(3) for _ in range(counts[i, j])
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])
```

[Run this code online](#)

```
cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().unstack().to_numpy()           # 3 x 3 table of cell means
n_ij = cells.size().unstack().to_numpy()
n, m = n_ij.sum(), (n_ij > 0).sum()
sse = ((trial["y"] - cells.transform("mean")) ** 2).sum()
s2 = sse / (n - m)                                # pooled within-cell variance
se_cell = np.sqrt(s2 / n_ij)
weighted_rows = (n_ij * ybar).sum(axis=1) / n_ij.sum(axis=1)  # = raw variety
                        averages
unweighted_rows = ybar.mean(axis=1)
se_weighted = np.sqrt(s2 / n_ij.sum(axis=1))
se_unweighted = np.sqrt(s2 * (1 / n_ij).sum(axis=1)) / 3
print(np.round(ybar, 3))
print("weighted  :", weighted_rows.round(3), " se", se_weighted.round(3))
print("unweighted:", unweighted_rows.round(3), " se", se_unweighted.round(3))
```

[Run this code online](#)

```
def hypothesis_test(L, ybar, n_ij, s2, dfe):
    """F test of  $L\mu = 0$  in the cell-means model, all cells occupied (cells in row-
    major order)."""
    u = L @ ybar.ravel()
    ss = u @ np.linalg.solve(L @ np.diag(1 / n_ij.ravel()) @ L.T, u)
    q = np.linalg.matrix_rank(L)
    F = ss / q / s2
    return ss, q, F, stats.f.sf(F, q, dfe)

C3 = np.array([[1.0, 0.0], [0.0, 1.0], [-1.0, -1.0]])  # any 3 x 2 matrix of
                        contrasts will do
avg3 = np.ones((1, 3)) / 3
L_A = np.kron(C3.T, avg3)                            # unweighted variety means equal
L_B = np.kron(avg3, C3.T)                            # unweighted site means equal
L_AB = np.kron(C3.T, C3.T)                          # all interaction contrasts zero
for name, L in [("variety", L_A), ("site", L_B), ("interaction", L_AB)]:
    ss, q, F, p = hypothesis_test(L, ybar, n_ij, s2, n - m)
    print(f"{name:12s} SS = {ss:6.3f} df = {q} F = {F:6.2f} p = {p:.4f}")
```

[Run this code online](#)

The script also checks that the variety sum of squares 4.958 is the increase in residual sum of squares when the cell means are fitted subject to H_A (Theorem 11.3.2(b)), and that a Helmert-type contrast matrix gives the same test. With one factor, counts only decide which contrasts split the between-groups sum of squares orthogonally (Theorem 15.4.1). With several they decide which hypotheses the printed sums of squares address.

17.1.5 Exercises

17.1.5.1 A. Check your understanding

Exercise. A1

A 2×3 layout has counts $n_{11} = 4$, $n_{12} = 0$, $n_{13} = 2$, $n_{21} = 3$, $n_{22} = 5$, $n_{23} = 1$. Give m , n , the rank of $\mathbf{W}$ and the error degrees of freedom of the cell-means model. Which of $\mu_{11} - \mu_{21}$, $\bar{\mu}_1 - \bar{\mu}_2$ and $(\mu_{11} + \mu_{13})/2 - (\mu_{21} + \mu_{23})/2$ are estimable?

Exercise. A2

In [Example \(Three varieties at three sites\)](#), compute from the counts alone the ratio of the standard errors of the unweighted and weighted means of V3, and check it against the quoted standard errors.

17.1.5.2 B. Practice**Exercise. B1**

Show that the F statistic of [Eq. 17.1.1](#) for a single contrast $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$ with all cells occupied is the one-way contrast statistic of Proposition 15.2.1, applied to the one-way layout whose groups are the cells. Which t statistic is it the square of, and what is its distribution under the hypothesis?

Solution. With $q = 1$, $\mathbf{L} = \boldsymbol{\lambda}^\top$ and $\mathbf{d} = 0$, [Eq. 17.1.1](#) is $F = (\sum_\omega \lambda_\omega \bar{y}_\omega)^2 / (s^2 \sum_\omega \lambda_\omega^2 / n_\omega)$. This is the contrast sum of squares $(\sum \lambda_\omega \bar{y}_\omega)^2 / \sum (\lambda_\omega^2 / n_\omega)$ of the one-way layout with groups ω , divided by the one-way error mean square, which is the pooled within-cell variance. Its signed root is $t = \boldsymbol{\lambda}^\top \bar{\mathbf{y}} / (s \sqrt{\sum \lambda_\omega^2 / n_\omega})$, which has a $t(n-m)$ distribution under the hypothesis. The factorial labels of the cells play no role.

Exercise. B2

Let $\mathbf{C}_a$ and $\mathbf{C}'_a$ be two $a \times (a-1)$ matrices whose columns are bases of $\mathbf{1}_a^\perp$. Show that $\mathbf{C}'_a = \mathbf{C}_a \mathbf{T}$ for a nonsingular $\mathbf{T}$, and deduce that the hypotheses H_A and H_{AB} of [Eq. 17.1.2](#), and their F statistics, are the same for both.

Solution. Both column spaces equal $\mathbf{1}_a^\perp$, so each column of $\mathbf{C}'_a$ is a combination of the columns of $\mathbf{C}_a$: $\mathbf{C}'_a = \mathbf{C}_a \mathbf{T}$. The rank of $\mathbf{C}'_a$ is $a-1$, so $\mathbf{T}$ is nonsingular. By Proposition 1.10.1(a), $(\mathbf{C}'_a)^\top \otimes \mathbf{v}^\top = (\mathbf{T}^\top \otimes 1)(\mathbf{C}_a^\top \otimes \mathbf{v}^\top)$ for any row vector $\mathbf{v}^\top$, and similarly for $(\mathbf{C}'_a)^\top \otimes \mathbf{C}_b^\top$. So the new hypothesis matrix is $\mathbf{S}\mathbf{L}$ with $\mathbf{S}$ nonsingular. Then $\mathbf{S}\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$. In [Eq. 17.1.1](#),

$$(\mathbf{S}\mathbf{L}\bar{\mathbf{y}})^\top (\mathbf{S}\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top \mathbf{S}^\top)^{-1} \mathbf{S}\mathbf{L}\bar{\mathbf{y}} = (\mathbf{L}\bar{\mathbf{y}})^\top (\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top)^{-1} \mathbf{L}\bar{\mathbf{y}}.$$

Exercise. B3

In [Example \(Three varieties at three sites\)](#), suppose that sites like S1, S2 and S3 make up 50%, 30% and 20% of the region where the varieties would be grown. Write down the estimate of the region-weighted mean yield of V1 and its standard error, and do the same for

the region-weighted difference between V1 and V3. Compute them with the listing. Which of the three kinds of marginal mean in this section would you report to a grower in the region?

17.1.5.3 C. Going deeper

Exercise. C1

A row of a two-way layout will receive N observations, to be allocated as $n_{i1}, \dots, n_{ib}$, with the aim of estimating $\sum_j w_j \mu_{ij}$ for fixed weights $w_j > 0$ summing to one. Treating the n_{ij} as real numbers, show that the variance of $\sum_j w_j \bar{y}_{ij}$ is minimized by $n_{ij} = Nw_j$, with minimum σ^2/N . What does this say about how the counts should relate to the target of inference?

Solution. Minimize $\sum_j w_j^2/n_{ij}$ subject to $\sum_j n_{ij} = N$. By Cauchy–Schwarz, $1 = (\sum_j w_j)^2 = (\sum_j (w_j/\sqrt{n_{ij}})\sqrt{n_{ij}})^2 \leq N \sum_j w_j^2/n_{ij}$, with equality iff $n_{ij} \propto w_j$, that is, $n_{ij} = Nw_j$. So the variance is at least σ^2/N , with equality exactly there. When the counts are proportional to the weights of the target, the target is estimated by the raw row average, the weighted and target means coincide, and no precision is lost. A design should allocate observations in proportion to the target weights, which for an unweighted target means equal replication.

Exercise. C2

In an $a \times b \times d$ layout, where C has d levels and all cells are occupied, order the cells lexicographically. Write, as Kronecker products, hypothesis matrices for: (a) equality of the unweighted A means $\bar{\mu}_{i..}$; (b) absence of the AB interaction in the table $\bar{\mu}_{ij.}$ of means averaged over C ; (c) absence of the three-factor interaction (Proposition 16.4.2). Give their ranks.